MILITARY SPECIFICATION

FAN, CIRCULATING

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers fans of the desk, bracket, and multipurpose high velocity types.

1.2 Classification.

1.2.1 Types and sizes. Fans covered by this specification shall be of the following types and sizes as specified (see 6.2).

Type I - Desk bracket, oscillating.

Size A - 10 inch diameter blade assembly.
Size B - 12 inch diameter blade assembly.
Size C - 16 inch diameter blade assembly.

Class 1 - 115 volts, single phase, 60 hertz.
Class 2 - 115 volts, single phase, 50 hertz.
Class 3 - 220 volts, single phase, 60 hertz.
Class 4 - 220 volts, single phase, 50 hertz.

Type II - Bracket, nonoscillating.

Size A - 30 inch diameter blade assembly.

Class 1 - 115 volts, single phase, 60 hertz.
Class 2 - 115 volts, single phase, 50 hertz.
Class 3 - 220 volts, single phase, 60 hertz.
Class 4 - 220 volts, single phase, 50 hertz.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Type III - High velocity multipurpose.

Size - Size based on performance (see 3.5 and table II).

Class 1 - 115 volts, single phase, 60 hertz.
Class 2 - 115 volts, single phase, 50 hertz.
Class 3 - 220 volts, single phase, 60 hertz.
Class 4 - 220 volts, single phase, 50 hertz.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

J-C-175	- Cable Assembly, Power, Electrical (3-wire Grounding Type, For Use on 125 volt Equipment).
CC-M-1807	- Motors, Alternating Current Fractional and Integral Horsepower (500 H.P. and Smaller).
GG-P-455	- Plates and Foils, Photographic (Photo-sensitive-Anodizing Aluminum).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple Wall.

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MIL-P-116	- Preservation, Methods of.
MIL-P-514	- Plates, Identification, Instruction and Marking Blank.
MIL-T-704	- Treatment and Painting of Materiels.
MIL-A-8625	- Andoic Coatings, for Aluminum and Aluminum Alloys.

STANDARDS

FEDERAL

FED-STD-141	- Paint, Varnish, Lacquer, and Related Materials: Methods of Inspection, Sampling, and Testing.
FED-STD-151	- Metals, Test Methods.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1472	- Human Engineering Design Criteria For Military Systems, Equipment And Facilities.

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE INC., (ANSI)

ANSI-S1.4 - General Purpose Sound Level Meters.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B499 - Measurement of Coating Thickness by the Magnetic Method: Nonmagnetic Coatings on Magnetic Basis Metals.
B633 - Electrodeposited Coatings of Zinc on Iron and Steel.
D1730-67 - Preparation of Aluminum-Alloy Surfaces for Painting.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 NW 7th Street, Miami, FL 33125.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

Pub. No. MG 1 - Motors and Generators.

(Application for copies should be addressed to National Electrical Manufacturers Association, 155 East 44th Street, New York, N.Y. 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

UNDERWRITER'S LABORATORIES, INC. (UL)

Standard No. 507 - Standard for Electric Fans.
UL Subject 746 - Document "E" Guide to Requirements for Polymeric Materials Used as Electrical Insulation.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 1285 Walt Whitman Road, Melville, L.I., NY 11746; or 1655 Scott Blvd., Santa Clara, CA 95050.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The fans shall be of the desk, bracket, and high velocity types and of various sizes as specified (see 6.2). They shall consist of a motor, base, blade, guard, electric cord, speed control device, and other parts necessary to constitute a complete functional product.

3.1.1 General criteria. Unless otherwise specified herein all requirements and data shall be at the voltages specified in 1.2 and at 60 or 50 hertz.

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3.2 First article (preproduction model). The contractor shall furnish one or more fans as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods and choice of design detail will produce fans that comply with the requirements of this specification.

3.3 Fire and casualty hazards. Each contractor shall submit proof that the fans he proposes to supply under this specification conform to the requirements of UL Standard No. 507. Listing by Underwriters' Laboratories will be accepted as evidence that the product conforms to this requirement. In lieu of the Underwriters' Listing, the contractor may submit independent proof satisfactory to the contracting officer that the fans conform to applicable requirements of the published standard, including method of tests of UL Standard No. 507. Compliance with the above preliminary requirements in regard to fire, safety and casualty hazards does not absolve the contractor from complete compliance with the other requirements of this specification in order to secure acceptance of his material or equipment.

3.4 Material. All material used in the manufacture of fan component parts shall be new. Material shall be free of defects that affect the fan's appearance, serviceability, or safety.

3.4.1 Material deterioration and control. The fan shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the item may be exposed.

3.4.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.4.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make such information available, upon request, to the contracting officer or designated representative.

3.4.4 Plastic. Each type of plastic used in the fans construction shall be identified by the contractor. Plastic components other than slow burning or self extinguishing shall not be used in the fan's construction. When plastic is used as an enclosure for live electrical parts or to support same, it shall be of a self extinguishing type. Testing to determine plastic type shall be in accordance with 4.6.3.18.

3.5 Component parts.

3.5.1 General. The fan shall consist of a motor, base, blade, guard, electric cord, speed control device, and other parts necessary to constitute a complete

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functional product. Type I, size C fans and type II and III fans shall have a handle designed for ease of carrying. Type I, sizes A and B fans shall have handles if specified (see 6.2).

3.5.2 Motor. Unless otherwise specified (see 6.2), the fan motor shall be of a single phase, induction, squirrel cage, guarded machine design of the permanent-split capacitor, two-value capacitor, or shaded pole type meeting the requirements specified in CC-M-1807 for Type I, Class I, Style A, Kind 2 (Group C or D) or Kind 3, Enclosure 1, Design D motors. Motor shall have torque sufficient to start and operate fan at any specified speed control setting. Motor shall be capable of operating continuously under maximum load without exceeding the temperature rise specified in UL Standard No. 507.

3.5.2.1 Voltage and frequency variation. The fans shall be capable of start up at any speed setting and of operation when tested as specified in 4.6.3.4.

3.5.2.2 Insulation. Insulation shall be class A or B in accordance with NEMA Publication No. MG-1.

3.5.2.3 Bearings. Bearings shall be ball, sleeve, or spherical design and shall be of the maintenance free permanent lubrication type.

3.5.2.4 Duty. Motor shall be designed for continuous operation when employed as the drive source for the type fans described herein.

3.5.2.5 Electromagnetic interference suppression. When specified in the invitation for bids (see 6.2), motors shall be designed for suppression of electromagnetic interference in accordance with requirements of MIL-STD-461.

3.5.3 Base.

3.5.3.1 Types I and II. The base shall not crack, break, or become distorted to such degree that its safety or effectiveness as a base is impaired when tested as specified in 4.6.3.5. The bottom of type I fan base shall have replaceable protective material or at least four protective feet for vibration isolation and prevention of surface marring. The protective material or feet shall be installed in such manner as to preclude marring of finished surfaces when base of fan is dragged across such surfaces with fan vertical axis inclined 30 degrees or more with respect to horizontal. Type I and II base shall be provided with mounting facilities suitable for fixed mounting of fan on vertical or horizontal surfaces.

3.5.3.2 Type III. The base shall be constructed of at least 5/16-inch steel rod (1/4-inch support rods are acceptable) or 3/4-inch steel tubing and shall be formed so that it will not show permanent distortion when tested as specified in 4.6.3.5. The base shall be such as to effect stable operation of fan under all normal operating conditions. Suitable means shall be provided for fixed mounting of fan on a vertical surface. The bottom of fan base shall be provided with a suitable protective material to preclude marring of finished surfaces.

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3.5.4 Guard.

3.5.4.1 Types I and II. The guard shall enclose the entire face, perimeter and reverse side of fan blades and associated moving parts. All guard openings shall be of a size that precludes entrance of a 1/2-inch diameter sphere when the sphere is thrust into any opening(s) with a force of one pound. The guard shall consist of a front section and a rear section, designed so that when assembled to form a complete guard the sections are secured in equally spaced positions around their interface. The sections must be readily separable, without the use of special purpose tools, to permit access to the blades and associated moving parts. The assembled guard shall be securely fastened to the motor housing. The guard shall be designed to preclude vibration noise, particularly noise caused by any portion of the guard coming in contact with any other portion of the guard or fan. The guard shall clear the base adjusting and locking device when operated in any specified tilt position (see 3.5.9). The guard shall neither contact the blade assembly or other rotating components nor show permanent distortion when tested in accordance with 4.6.3.6.1.

3.5.4.2 Type III. The guard or protective housing shall enclose the entire face, perimeter and reverse side of fan blades and associated moving parts. All openings in guard or housing shall be of a size that precludes a 1/2-inch diameter sphere when the sphere is thrust with a force of one pound. The guard or protective housing shall be designed to permit ready access, without the use of special purpose tools, to the blade assembly and associated components for maintenance and cleaning. The guard or housing shall be designed to preclude vibration noise, particularly noise caused by any portion of the guard or housing coming in contact with any other portion of the guard housing or fan. The guard shall neither contact the blade assembly or other rotating components nor show permanent distortion when tested in accordance with 4.6.3.6.2.

3.5.5 Blade assembly. The blade assembly shall consist of a complete air screw having at least three blades together with a hub or central support. The fan shall be designed so that blade assembly will not contact the fan guard or motor housing should the blade assembly become loose on the motor shaft. The blade assembly shall not chip, crack, break, or show sign of distortion when operated at a speed of 150 percent of the maximum fan speed. The diameter of the type I fan blade assemblies shall be 10, 12, and 16 inches, respectively, for the size A, B and C fans and the diameter of the type II fan blade assembly shall be 30 inches, all diameter tolerances being $\pm$1/2 inch. The diameter of the type III fan blade assembly shall be as necessary to meet performance requirements provided maximum dimensions specified in 3.8 are not exceeded.

3.5.5.1 Fan blades. The fan blades shall be constructed of metal or plastic and shall be of such design that normal cleaning and dusting shall not cause bending out of alinement. Nonintegral metal blades shall be secured to the hub or central support with not less than two rivets per blade or other means of equal strength and rigidity. Other blade designs, if not integral parts of blade assembly shall be secured to hub or central support in a manner that provides

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strength and rigidity equivalent to that specified above for metal blades. Type I and III fan blades shall show no distortion or fracture and shall show no displacement with respect to the hub assembly upon application of 20 pound-feet of torque. Type II fan blades shall show no distortion or fracture and shall show no displacement with respect to hub assembly upon application of 40 pound-feet of torque.

3.5.5.2 Hub or central support. The hub or central support shall be readily replaceable on the motor shaft, forming a snug fit on the motor shaft and secured in a manner that precludes turning. The hub or central support of type I and III fan blades shall show no distortion or fracture and shall not separate from fan blades when subjected to an axial pull force of 25 pounds. The hub or central support of type II fans shall show no distortion or fracture and shall not separate from fan blades when subjected to an axial pull force of 30 pounds.

3.5.5.3 Blade assembly balance (applicable to type III fans only). The blade assembly on the type III fan shall be balanced to an accuracy of at least 3 gram-centimeters.

3.5.6 Speed regulating device.

3.5.6.1 Speed regulating device positions. The speed regulating device shall have not less than the number of speed positions indicated in table I and these positions shall be marked.

3.5.6.2 Speed regulating device low speed positions. The speed regulating devices shall be such that the low speed positions provide the percentage range of maximum speeds indicated in table I, when operating at rated voltage and frequency.

TABLE I. Speed regulating device positions.

Type	Size	Minimum number of positions	Percent of maximum fan speed with switch on lowest speed position
I	A	1	------
I	B	2	60 to 85 percent
I	C	3	60 to 85 percent
II	A	2	Not over 80 percent
III		2	Not over 80 percent

3.5.6.3 Continuous operation. No damage to fan or speed regulating device shall occur if speed regulating device is left continuously in any speed position.

3.5.6.4 Fire resistance. The speed regulating device shall be nonflammable and designed to permit continuous operation at any speed setting.

3.5.6.5 Voltage variation. The fan shall be capable of starting at any speed setting at a voltage 10 percent lower than rated voltage.

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3.5.6.6 Speed regulating device location. With the exception of type I, size A fans, type I and II fan speed regulating device mechanisms (except control means) and associated wiring shall be completely enclosed within the fan base with access provided via detachable base plate cover, preferably underneath the base. The speed regulating control shall be positioned in such location on the fan base so as to be easily accessible when fan is used in either desk or bracket positions. All components of the type III speed regulating device shall be enclosed except the control means (handle, button, or knob).

3.5.6.7 Clearance speed regulating device. Types I and II fans shall have a clearance of not less than 1-1/2 inches between the speed regulating device control means and fan guard for all positions of the fan. All fan types shall have a clearance of not less than 2 inches between the control means and fan blade for all positions of the fan.

3.5.7 Electric cord and plug. Unless otherwise specified (see 6.2) the cable assembly shall be any class and grade specified in J-C-175. The cable assembly shall be at least 10 feet from plug connector to the point of entrance into the fan. If cable assembly is to be of a specific color, such color shall be specified (see 6.2), otherwise color shall be compatible with color of fan. On type I and III fans the cable assembly shall enter fan base and be concealed.

3.5.7.1 Oscillator cord (type I fans only). The oscillator cord shall be of flexible stranding and shall be no greater than 3/8 inch in overall diameter. The cord shall not enter the motor through the motor end adjacent to blade assembly. Cord shall have sufficient length between motor and stationary base to insure freedom of oscillatory movement.

3.5.8 Oscillatory mechanisms (type I fans only). The angle of oscillation shall be not less than 55 degrees for the type I, size A fan, and not less than 75 degrees for the type I, sizes B and C fans. Oscillating gears and drive and train shall be designed to provide a smooth uninterrupted movement of the main fan blade assembly during each oscillatory cycle. Metal oscillating gears shall be lubricated and contained within a sealed gear box to prevent loss of lubricant and collection of abrasive matter. Oscillating gears of the self lubricating type (nylon, delrin, etc.) shall be enclosed to preclude dust and other abrasive matter. Oscillatory mechanism shall be of design that can be disengaged in any position within specified oscillation limits, whereupon the fan shall remain stationary in the position of disengagement.

3.5.9 Tilt adjustment and locking devices. Tilt adjustment and locking devices shall be of sturdy construction, remaining secure when tested in accordance with 4.6.3.9. Adjustment and locking devices shall be designed to permit adjustment and locking in any specified tilt positon without the use of tools. If a wing nut is used to secure the locking device, it shall have a smooth contoured surface and shall be at least 1-inch in diameter across the "wings". If a knob is used, it shall also be at least 1-inch in diameter. Should the locking device be of a design that would permit the fan to become accidentally disassembled, upon loss of securing lock (wing nut, knob, etc.), a

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safety stop shall be provided to preclude accidental disassembly. Tilt adjusting devices shall be designed so that tilt adjustments can be accomplished with ease without necessity of physically supporting the total weight of the portion of the fan being adjusted, without binding, and without marring fan protective finishes. Type I fans shall be designed so that assembly shall not be required in order to utilize in either desk or bracket positions. Tilt adjustment on type I fans shall permit operation within the range of at least 10 degrees below horizontal to at least 10 degrees above horizontal when used as a desk fan, and from horizontal to at least 25 degrees below horizontal when used in the bracket position. Tilt adjustment on type II fans shall permit operation within the range from horizontal to at least 25 degrees below horizontal. Tilt adjustment on type III fans shall permit operation through a range of 180 degrees, from straight up to straight down. The above tilt angles shall be construed as angles formed by motor shaft axis and the horizontal plane.

3.6 Fan performance. Unless otherwise indicated, fan performance, listed in table II, shall be determined at rated voltage and frequency at normal operating temperature of the motor in the high speed position. Type III fans shall be provided with vanes, louvers, or other means as necessary to impart an essentially linear jet stream character to the air movement. Thrust per watt ratios shall be obtained by dividing measured thrust (in pounds) by the measured fan motor power input (in watts) required to produce the particular measured thrust.

TABLE II. Fan performance requirements.

Type	Size	Maximum power input (watts)	Minimum thrust/watt ratio (lbs./watt)	Minimum thrust (lbs.)	Minimum CFM air delivery 9 ft. in front of blade	Maximum A weighted noise level (decibels)
I	A		.0038	.20		49
	B		.0061	.35		51
	C		.0128	.83		57
II	A		.0114	4.2		75
III		145			2500	60

3.6.1 Vibration limits (applicable to type I and II fans only). When tested in accordance with 4.6.3.16, the rotating components of the type I fan shall be balanced to the extent that total vibration shall not exceed 0.003 times the distance in inches between an upward extension of a vertical line running through the approximate center of the motor support and point of measurement. The distance is to be measured parallel to the motor axis with motor housing adjusted for zero degrees tilt. The rotating components of the type II fan shall be balanced to the extent that total vibration, when measured as outlined above, shall not exceed 0.009 times the distance in inches between an upward extension of a vertical line running through the approximate center at the motor support and point of measurement.

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3.7 Finish. Unless otherwise specified (see 6.2), the finish shall be the manufacturer's standard practice provided that the color of the fan is not predominately black or gray and provided that the finish is suitable for business and home use. When specified (see 6.2), finish shall be as follows:

3.7.1 Aluminum alloy parts. Aluminum parts shall be protected for corrosion resistance by either anodizing, as per MIL-A-8625 Type II, Class 1 to a minimum thickness at 0.0007 inch or covering with enamel or lacquer coating.

3.7.2 Ferrous parts. Ferrous parts shall be protected for corrosion resistance by either chromium or zinc plating or covering with enamel, or lacquer coating. No rust shall appear, except as noted, when tested in accordance with 4.6.3.17.3. All ferrous parts which are exposed to the outside environment shall be tested.

3.7.3 Coatings.

3.7.3.1 Enamel and lacquer coatings. Prior to coating, ferrous parts shall be treated in accordance with manufacturer's corrosion inhibitive methods provided these methods are at least equivalent to these applicable as stated in MIL-T-704. Aluminum parts shall be pretreated in accordance with one of the methods outlined in ASTM D1730. The dry film thickness shall be between .7 and 1 mil. and shall have a dry film contrast ratio of not less than 0.98. Coatings when applied at the above thickness shall ribbon or curl without flaking or otherwise loosening from the primer or base metal. Coating flexibility shall be such that no cracking or crazing occurs when tested in accordance with 4.6.3.17.1.

3.7.3.2 Chromium or zinc plating. Chromium plated finishes shall have a polished appearance. Minimum thickness of the total plating shall be 0.0004 (0.4 mils) when tested in accordance with 4.6.3.17.2. Zinc plated finishes shall have a polished appearance and shall be as per ASTM B633 SC3, Finish Type I.

3.8 Maximum dimensions, type III fan. The dimensions of type III fan, shall not exceed 19 inches high, 18 inches wide, and 11 inches deep.

3.9 Human factors. The fan shall conform to the applicable human factors engineering criteria as specified in MIL-STD-1472.

3.10 Identification marking. The fan shall be identified in accordance with MIL-STD-130. The marking shall be applied to the fan on plates conforming to MIL-P-514, Type I, Style 3, Composition A, Class 2 or Composition C (GG-P-455, Type I or II, Grade A, Class 1). Plates shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible. The plate shall contain at a minimum, contract number, date of manufacture, voltage and frequency.

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3.11 Workmanship. The finished fan shall be clean and conform to the quality and grade of the products established by this specification. The fan shall be free from any workmanship or design defects that affect appearance and serviceability and any workmanship or design defects that may cause injury to the user.

3.11.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.11.2 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.11.3 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

3.11.4 Welders and welding.

3.11.4.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1 - Structural Welding Code, Section 5, Qualifications.
ASME - Boiler and Pressure Vessel Code, Section IX, Welding Qualification.

Contractors who only make horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, contractor's standard welder qualification may be substituted in lieu of the above codes. The contractor shall be responsible for determining that automatic welder equipment operators are capable of producing quality welds.

3.11.4.2 Welding. The surface of parts to be welded shall be free from oxide, scale, paint, grease, oil, or any other foreign matter. Immediately before welding, oxide shall be removed by an abrading or a chemical process. Grease and oil shall be removed by wiping with a solvent such as acetone. All surfaces shall be dried after cleaning. Welds shall transmit stress without permanent deformation or failure when the parts connected by welds are subjected

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to proof and service loadings. Complete and uniform penetration and fusion of the metals shall be obtained on all welds. All exposed welds or seams shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding whenever possible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1, the contractor is responsible for insuring that components and materials used were manufactured, tested and inspected in accordance with the specified requirements of referenced subsidiary specifications and standards, or if none, in accordance with this specification.

4.2 Code and standards compliance. The contractor shall provide the Government representative with proof of compliance with the Underwriters' Laboratories requirements of 3.3.

4.3 Classification of inspections. The inspection requirements are classified as follows:

a. First Article (preproduction) inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection.

4.4.1 Examination. The first article (preproduction) fans shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. The first article (preproduction) fans shall be subjected to the tests marked "X" in column 2 of table III in the sequence listed, or in the sequence listed in a Government approved test plan. Failure of any test shall be cause for rejection.

4.5 Quality conformance inspection.

4.5.1 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be made in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

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4.5.1.1 End product inspection. All fans of the same type and size offered for acceptance at one time shall be considered a lot for the purpose of inspection. The sample unit for these inspections shall be one completely fabricated fan.

4.5.1.1.1 Visual examination. The fans shall be visually examined as specified in 4.6.1. Defects shall be classified as follows. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 percent defective for major detects and 6.5 percent defective for minor detects.

4.5.1.1.2 Dimensional examination. Examination shall be made of the end item to determine compliance with dimensional requirements. Any noncompliance with specified requirement shall constitute a defect. The inspection level shall be S-2 and the AQL shall be 6.5 percent defective for all defects.

4.5.1.1.3 Tests. The fans shall be subjected to the tests marked "X" in column 3 of table III in the sequence listed, or in the sequence listed in a Government approved test plan. The tests shall be conducted using inspection level S-2 with an AQL of 4.0 percent detective for all defects.

4.6 Inspection procedure.

4.6.1 Examination. The fans shall be examined for the following defects:

Number	Requirement	Defect	Major	Minor
101	Materials deterioration	Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.	X	
102	Dissimilar metals	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	X	
103	Material documentation	Contractor does not have documentation available for identification of material, material finishes or treatments.	X	
104	Finish	Not finished as required.	X	
105	Design	Any characteristic not in accordance with the specified requirements.	X	
106	Construction and workmanship	Fractured, split, bowed, malformed, or otherwise impaired.	X	
107	General (applicable to all components and assemblies)	Sharp burrs, slivers, or splinters that may be injurious to personnel.	X	
201	Assembly	Not properly assembled or secured.		X
202	Marking identification	Missing, incomplete, not legible, not as specified.		X

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4.6.2 Inspection schedule. The inspection schedule is contained in table III.

TABLE III. Inspection Schedule. 1/

Type of Inspection			Examination or Test	Insp. Par	Req. Par.
Insp. Req.	First article	Qual. Conf.			
1	X	X	Examination	4.6.1	3.1, 3.5
2	X	X	Motor	4.6.3.2	3.5.2
3	X		Electromagnetic Interference	4.6.3.3	3.5.2.5
4	X		Voltage & Frequency Variation	4.6.3.4	3.5.2.1
5	X		Distortion	4.6.3.5	3.5.3.1, 3.5.3.2
6	X		Guard	4.6.3.6	3.5.4.1, 3.5.4.2
7	X		Blade Torque	4.6.3.7	3.5.5.1
8	X		Hub Axial Pull Force	4.6.3.8	3.5.5.2
9	X	X	Adjustment & Locking Devices	4.6.3.9	3.5.9
10	X	X	Overspeed	4.6.3.10	3.5.5
11	X	X	Speed	4.6.3.11	3.5.6.2
12	X	X	Thrust	4.6.3.12	3.6, Table II
13	X	X	Power Input	4.6.3.13	3.6, Table II
14	X		Noise	4.6.3.14	3.6, Table II
15	X	X	Air Delivery	4.6.3.15	3.6, Table II
16	X		Vibration	4.6.3.16	3.6.1
17	X	X	Finish	4.6.3.17	3.7
18	X		Burning Rate/Self Extinguishing	4.6.3.18	3.4.4
19	X		Human Factors	4.6.3.19	3.9

1/ Perform inspection marked "X".

4.6.3 Tests.

4.6.3.1 Conditions. Unless otherwise specified herein, tests shall be conducted at rated voltage and employing 60 hertz electrical power or 50 hertz electrical power, as applicable. The frequency tolerance for all tests shall be plus or minus 1 percent. Unless otherwise specified herein, no braces, sealing materials, tiedown devices, or other items that are not inherently a part of the operating fan shall be employed. Unless otherwise specified herein, no alinement of components, or other servicing shall be performed on the first article or production fans after testing of the fan has been initiated. Components

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subject to atmospheric corrosion shall not be treated with any treatment such as grease or oil in order to meet test requirements.

4.6.3.2 Motor. With exception of temperature rise test, motor tests shall be conducted in accordance with requirements of CC-M-1807. Temperature rise test shall be in accordance with UL Standard No. 507. Failure of above tests shall be cause for rejection of product.

4.6.3.3 Electromagnetic interference. Electromagnetic interference suppression tests shall be conducted in accordance with MIL-STD-461. The contractor shall furnish the contracting officer a test plan and a test report as required by MIL-STD-461. Disapproval of the test report shall constitute failure of this test.

4.6.3.4 Voltage and frequency variation. Subject the fan to each of the voltage and frequency conditions contained in table IV. Nonconformance to 3.5.2.1 shall constitute failure of this test.

TABLE IV. Voltage-frequency variation.

Test	Voltage	Frequency	Voltage (Tolerance ±1%)		Frequency	
	Rated	Rated	115	220	60	50
a	High	High	126.5	242	63	52.5
b	Low	Low	103.5	198	57	47.5
c	Low	High	103.5	198	63	52.5
d	High	Low	126.5	242	57	47.5

4.6.3.5 Distortion.

4.6.3.5.1 Type I and II fans. Type I and II fans shall be dropped free fall from a height of 15 inches onto a concrete or rigid metal surface so as to strike on one edge of base. Subsequently base shall be examined to determine compliance with 3.5.3.1. Noncompliance with 3.5.3.1 shall be construed as test failure and shall be cause for rejection. Damage to other parts of fan shall not be considered.

4.6.3.5.2 Type III fans. The type III fan shall be placed in normal operating position and subjected to a vertical force of 50 pounds for a period of ten minutes. Subsequently, the fan shall be examined to determine if permanent distortion of base has occured. Evidence of permanent base distortion shall be construed as test failure and shall be cause for rejection.

4.6.3.6 Guard. To determine compliance with 3.5.4.1 and 3.5.4.2 the following tests shall be performed. A 1/2-inch diameter sphere shall be thrust into any selected guard opening(s) with a force of one pound from a direction and in a manner most apt to favor penetration of the guard. Sphere penetration of the guard shall constitute test failure and be cause for rejection of fan.

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4.6.3.6.1 Type I and II fans. Type I and II fan motor housings with blade assembly and guard shall be held securely in a position such that the plane of blade rotation is horizontal with the guard facing upward. The fan guard when in correct test position shall be supported only by means of attachment to motor housing incorporated in fan design. No supports shall be placed underneath guard during test. After positioning, a round steel shaft 4 inches in diameter and weighing 50 pounds shall be placed so that the shaft exerts a force of 50 pounds over randomly selected areas of the fan guard face, one of which shall be the center portion of fan face. With each placement of the shaft, the fan blade assembly shall be rotated by hand whereupon the guard shall not contact the fan blades or other rotating components. Blade assembly contact, other rotating component contact, permanent guard distortion at any time during testing or non-compliance with 3.5.4.1 shall be construed as test failure and shall be cause for rejection.

4.6.3.6.2 Type III fans. The type III fan shall be adjusted for a tilt position such that airflow, if in operation, would be straight up (tilt position of 90 degrees above horizontal) with plane of blade rotation horizontal. After positioning, a round steel shaft 4 inches in diameter and weighing 50 pounds shall be placed so that the shaft exerts a force of 50 pounds over randomly selected areas of the fan guard face, one of which shall be the center portion of fan face. With each placement of the shaft, the fan blade assembly shall be rotated by hand whereupon a guard shall not contact the fan blades or other rotating components. Repeat the above procedure with fan tilt adjusted so that air flow, if in operation, would be straight down (a tilt position of 90 degrees below horizontal) with plane of blade rotation horizontal. When in position, the 50 pound shaft shall be placed over randomly selected areas at the rear of the fan guard and the blade assembly again rotated by hand. Blade assembly contact, other rotating component contact, permanent guard distortion at any time during testing, or noncompliance with 3.5.4.2 shall be construed as test failure and shall be cause for rejection.

4.6.3.7 Blade torque. Each blade of the blade assembly shall be subjected to a torque of 20-pound-feet or 40-pound-feet, as applicable dependent upon fan type (see 3.5.5.1) applied in such manner that the force causing the torque is directed in the plane of blade rotation. Torque shall be applied for one minute whereupon blades shall be examined for displacement from the hub assembly. Evidence of displacement, distortion, fracture, or noncompliance with 3.5.5.1 shall be construed as test failure and shall be cause for rejection.

4.6.3.8 Hub axial pull force. The blades of the complete assembly shall be held in a fixed position whereupon an axial pull force of 25 pounds or 50 pounds, as applicable dependent upon fan types (see 3.5.5.2) shall be applied to the hub or central support for one minute alternately in both axial directions. After force application the hub or central support shall be examined for evidence of displacement distortion or fracture. Evidence of displacement, distortion, fracture, or noncompliance with 3.5.5.2 shall be construed as test failure and shall be cause for rejection.

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4.6.3.9 Adjustment and locking devices. Prior to test, adjust type I and III fans for a tilt setting of 10 degrees above horizontal and type II fan for a horizontal tilt setting and lock in position. Subsequently, the complete fan shall be raised 6 inches above a concrete or rigid metal surface and allowed to drop free fall. Prior to release, the bottom of the fan base shall be positioned so that it is parallel to the rigid impact surface. Repeat the drop two additional times, then examine for compliance with 3.5.9. Tilt settings after completion of three drops shall not vary by more than 2 degrees from the settings established prior to test. Evidence of variation shall be construed as test failure and shall be cause for rejection.

4.6.3.10 Overspeed. The blade assembly shall be removed from the motor shaft and secured to the shaft of a variable-speed motor having a high speed rating of at least 175 percent of the fan high speed rating. The blade assembly shall then be run at a speed of 150 ±3 percent of the maximum speed of the fan for one hour. Upon completion of the test, the blade assembly shall be examined for cracks, breaks, chipping, and tightness of blades to hub or central support. The speed of rotation shall be measured stroboscopically or with a counter that imposes negligible load on the motor. Cracks, breaks, chipping or signs of distortion shall be construed as test failure and shall be cause for rejection.

4.6.3.11 Speed. The fan shall be operated at both low and maximum speed settings and speed (r.p.m.) shall be measured stroboscopically or with a counter that imposes no appreciable load on fan motor. The sensitivity and accuracy of the test apparatus shall be such that differences of 20 r.p.m. may be detected. Inability to meet requirements of 3.5.6.2 shall be construed as test failure and shall be cause for rejection.

4.6.3.12 Thrust.

4.6.3.12.1 Sensitivity of apparatus. The thrust shall be measured by means of an apparatus having a sensitivity such that a change in thrust of 0.01 pounds causes a motion of the pointer or indicating device of at least 1/8 inch.

4.6.3.12.2 Description of apparatus. The apparatus shall be calibrated by means of standard weights. In case of controversy, the apparatus developed at the National Bureau of Standards shall be used. This apparatus is based on the principle of a pendulum, counterbalanced to give adequate sensitivity. A platform is suspended by four rods of equal length, the connection of the rods to the platform and to the supporting framework being through ball bearings. The platform is thus free to move back parallel to a horizontal plane. The weight of the platform and a part of the weight of any fan placed on the platform are counterbalanced by two counterweights placed above the plane of the four upper supporting bearings. In practice, the overhead weights are adjusted to give an overbalanced condition with no fan; and the desired stability and sensitivity are obtained by adding weights to the platform. The current for the fan motor is brought in through the frame and through an insulated line carried through insulated brush contacts. A scale graduated in terms of the tangent of the deflection angle is attached to the platform.

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4.6.3.12.3 Calibration of apparatus. The scale is calibrated and the sensitivity measured by applying known horizontal forces by means of a light thread running over a ball bearing pulley to standard weights.

4.6.3.12.4 Alternate method for determining thrust. In lieu of determining thrust by the above method, the following alternate method may be used. Using the apparatus described in 4.6.3.12.2, with exception of graduated scale pointer device, place the fan on the platform with the motor axis in a horizontal plane and mark the platform's initial position with a stationary marker. After marking position of the platform, attach a light thread to center of the fan guard. Run thread over a pulley and down to a position below the platform. At this point, attach a bucket containing lead weights to the thread. Turn fan on and set to highest speed position. Change weight in bucket until platform is returned to the initial position as marked above. Detach bucket and weigh on a scale or balance having an accuracy of at least 0.001 pounds. The weight of the bucket and lead weights, so obtained, gives the direct measurement of fan thrust.

4.6.3.12.5 Basis of failure. Noncompliance with the thrust requirements of 3.6 and table II shall be construed as test failure and shall be cause for rejection.

4.6.3.13 Power input. The sensitivity and accuracy of the instrumentation to measure power input shall be such that differences of one watt may be detected. Inability of the fan to meet the power requirements of 3.6 and table II shall be construed as test failure and shall be cause for rejection.

4.6.3.14 Noise.

4.6.3.14.1 Instrumentation. Noise measurement shall be made with a general purpose type 2 sound level meter conforming to ANSI Standard S1.4. The sound level meter shall conform to response curve A of the above ANSI S1.4 Standard.

4.6.3.14.2 Test room. The test room for measuring the noise shall be so constructed that the noise reflected from the walls, floor and ceiling will have a negligible effect on the measurements taken. The size of the test room shall be such that recirculating air currents will not affect the motor speed or the power input. The noise level of the test room when the fan is not in operation shall be at least 6 decibels lower than the lowest reading which is obtained during the operation of the fan.

4.6.3.14.3 Procedure. The fan shall be mounted in its normal operating position centrally in the test room, and the motor axis of rotation shall be horizontal and at least 3 feet above the floor. Resilient means shall be interposed between the fan and its support so that the transmission of vibration to the support and the reaction of the fan are both negligible. Seven measurements shall be taken at stations in one horizontal plane which includes the motor axis of rotation. These stations shall all be on a circle having a 3-foot radius, measured from the center of the hub of the blade assembly, and shall be spaced equidistance at 45 degrees position on this circle, including one positon on the fan axis of rotation at the back of the blades and omitting

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the position on the fan axis of rotation at the front of the blades. The fan speed at the beginning of the noise test shall be kept constant during the test. The average of the readings shall be taken at the noise level of the fan at the test speed. If the range of the sound level measurement is less than 10 decibels, their average can be obtained by an arithmetical average of the decibel readings. However, the average of a number of sound level readings is defined as 10 times the logarithm to the base 10 of the average of the antilogarithms of 1/10 of the separate decibel values; this value to be used where differences in reading between positions exceed 10 decibels. The following example will illustrate the method of obtaining the average:

Microphone position	Sound level	Antilogarithm to the base 10 of 1/10 of the decibel sound level
	Decibels	
A	60	1,000,000
B	50	100,000
C	40	10,000
	Total.....	1,110,000
	Average.....	370,000

Log10 370,000 = 5.57 10 Log10 370,000 = 55.7 or the average decibel value

If the noise level of the room when the fan is not in operation is less than 10 decibels lower than any reading taken with the fan in operation, the reading shall be corrected in accordance with the following:

Difference in decibels between room level and fan level reading.	Subtract this decibel correction from fan level reading.
6	1.3
7	1.0
8	0.8
9	0.6
10	0.4

4.6.3.14.4 Basis of failure. Noncompliance with paragraph 3.6 and table II shall be construed as test failure and shall be cause for rejection.

4.6.3.15 Air delivery. Air delivery shall be measured in cubic feet per minute and applies to type III fans only.

4.6.3.15.1 Air flow measuring instrument. The air flow shall be measured with an anemometer(s) of the commercial biram or paddlewheel type having a minimum diameter of 2-1/2 to 3 inches and an air-speed range of 50 to 1500 feet per minute.

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4.6.3.15.2 Air flow test room. The test room for measuring the air flow shall be so constructed that all wall openings, including doors and windows, can be tightly closed. The test room shall be large enough to prevent disturbance of the normal air stream pattern during the test. The minimum size of the unobstructed test space within the test room shall be 15 feet wide, 30 feet long and 9 feet high.

4.6.3.15.3 Anemometer stand. The stand that supports the anemometer shall offer minimum resistance to the flow of air. It shall be so constructed that it will support the anemometer in any one of the stations required for any complete air delivery test. The anemometer clamp shall present minimum obstruction to the normal air flow. The operator shall be able to change the position of the anemometer without disturbing the normal air stream pattern.

4.6.3.15.4 Air delivery measurement. The fan shall be mounted so that its axis of rotation is horizontal, parallel to the long side of the test room, 4 feet above the floor and centered in the unobstructed width of the test space. The plane of rotation of the fan blades shall be 5 feet from the back wall. The anemometer shall be rigidly supported in the air stream with its axis of rotation parallel to the axis of rotation of the fan. The anemometer stand shall be so located that the center of the anemometer, in all stations, will be in a single vertical plane which is parallel to and 9 feet in front of the plane of rotation of the fan blades. Test readings shall be taken at 2 inch intervals over a four foot span, both vertical and horizontal lines at a distance of 9 feet in front of the plane of rotation of the fan blades. The center of the four foot span referenced above shall be on a line coincident with the axis of rotation of the fan blades. A total of 49 test readings shall be taken. Each test reading shall be taken over an operating period of not less than two minutes. All readings shall be taken at maximum fan speed. The air delivery in cubic feet per minute shall be the average of all test readings in feet per minute multiplied by the area of a 4 foot diameter circle (12.56 square feet).

4.6.3.15.5 Basis of failure. Noncompliance with 3.6 and table II shall be construed as test failure and shall be cause for rejection.

4.6.3.16 Vibration. The vibration test applies to type I and II fans only. Vibration shall be measured in a plane perpendicular to the motor shaft axis at the rear of the fan motor housing or cover. Measurements shall be taken at each specified speed setting (see table I) with fan operating at rated voltage and frequency.

4.6.3.16.1 Procedure. Test procedure is as follows:

a. Place assembled fan in a normal upright position and set tilt to 0 degrees. Oscillating fans shall be locked in a stationary (non-oscillating) position at center of oscillatory swing.
b. Using a prick punch, make a pin point indentation (point of measurement) in the approximate center of rear motor housing/cover.

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c. Using a microscopic device mounted on an independent stable base and having a reticulated eyepiece, focus on the above indentation. Calibrate eyepiece so that each minor division is in the range of.001 - .005 inches as applicable. When focusing insure that sufficient light is available.
d. Place fan in operation and observe that the indentation, when viewed through the microscopic device, appears to inscribe a straight line. Measure and record the length of the inscribed line. This length will be the total vibration movement.

4.6.3.16.2 Basis of failure. Vibration in excess of the limits specified in 3.6.1 shall be construed as test failure and shall be cause for rejection.

4.6.3.17 Finish.

4.6.3.17.1 Enamel and lacquer. Dry film thickness and contrast ratio shall be determined in accordance with test methods 6183 and 4122.1 respectively of FED-STD-141. Ability to ribbon or curl without flaking or loosening shall be determined in accordance with Method 6304.1 of FED-STD-141. Compliance with the flexibility of 3.7.3.1 shall be determined by rapidly doubling a coated tinplate test panel over a 1/8-inch steel rod mandrel in accordance with Method 6221 of FED-STD-141. Prior to doubling, the panel shall be: (1) allowed to air dry for 2 hours at a temperature of 23° ±1.1° C (73.4° ±2° F) and a relative humidity of 50 ±4 percent; (2) baked for 3 hours at a temperature of 105° C; and (3) conditioned for at least 1/2 hour at a temperature of 23° ±1.1° C (73.4 ±2° F) and a relative humidity of 50 ±4 percent. Noncompliance with 3.7.3.1 shall be construed as test failure and shall be cause for rejection.

4.6.3.17.2 Thickness of chromium and zinc finishes. Determine total thickness of chromium plated surfaces, and zinc plated surfaces by either the magnetic test of ASTM B499 or the electronic test, method 520.1 of FED-STD-151. Noncompliance with 3.7.3.2 shall be construed as test failure and shall be cause for rejection.

4.6.3.17.3 Resistance to rust test. All grease shall be removed from the parts to be tested by immersion in 1,1,1-trichloroethane for 10 minutes. The parts are then immersed for 10 minutes in a 10 percent solution of ammonium chloride in water at a temperature of 20 ±5° C. Without drying, but after shaking off any drops, the parts are placed for 10 minutes in a box containing air saturated with moisture at a temperature of 20 ±5° C. After the parts have been dried for 10 minutes in a heating cabinet at a temperature of 100 ±5° C, their surfaces shall show no signs of rust. Any evidence of rust, with the exception of traces on sharp edges and any yellowish film removable by rubbing shall be construed as test failure and shall be cause for rejection.

4.6.3.18 Burning rate/self extinguishing test. Testing to determine plastic type, either slow burning or self extinguishing, shall be in accordance with Underwriters' Laboratories, Inc., "Burning Test For Polymeric Materials Classed As Slow Burning", and "Burning Test For Polymeric Materials Classed As

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Self-Extinguishing", reference pages E3, E4, and E5 of UL subject 746 of March 1, 1967. Failure to meet requirements of 3.4.4 as determined per this test shall be cause for rejection.

4.6.3.19 Human factors. The fan shall be evaluated to determine that the human factors criteria have been met. Nonconformance to 3.9 shall constitute failure of this test.

4.7 Inspection of packaging.

4.7.1 Quality conformance inspection of pack.

4.7.1.1 Unit of product. For purposes of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.1.3 Examination. Samples selected in accordance with 4.7.1.2 shall be examined for the following defects. The AQL shall be 2.5 percent defective.

108.	Materials, methods, and containers not as specified for level A or B. Each incorrect material, method or container shall constitute one defect.	
109.	Fan not immobilized to prevent movement or damage as specified for level A or B.	
110.	Quantities packed together exceed the limitations of the applicable box for level A or B.	
111.	Preservation and packing not in accordance with the referenced document as specified for commercial.	
112.	Marking missing, illegible, incorrect, or incomplete for level A, B or commercial.	

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or commercial as specified (see 6.2).

5.1.1 Level A. Each fan shall be preserved in accordance with MIL-P-116, Method IIb. The interior container shall be the contractors standard commercial carton or a close-fitting box conforming to PPP-B-636, class domestic, variety SW, grade as applicable to the weight, style optional. The exterior container shall conform to PPP-B-636, class weather-resistant, variety and grade as applicable to the weight of contents, style optional. The fan shall be blocked, braced, or cushioned as applicable within the interior container to prevent movement or damage.

5.1.2 Level B. Each fan shall be preserved as specified in 5.1.1 for level A, except the method of preservation shall be IA-14.

5.1.3 Commercial. Each fan shall be preserved in accordance with MIL-STD-1188.

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5.2 Packing. Packing shall be level A, level B or commercial, as specified (see 6.2).

5.2.1 Level A. Fans, preserved as specified in 5.1 shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional in quantities not to exceed the weight limitations of the box. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. Fans, preserved as specified in 5.1, shall be packed together as specified in 5.2.1 for level A, except the boxes shall be domestic type, or the fans shall be packed in boxes conforming to PPP-B-636, V3c, style optional or in boxes conforming to PPP-B-640, class 2, style optional. Fans packed in fiberboard boxes shall not exceed the size and weight limitations of the applicable box. Box closure and sealing shall be in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. Fans, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The fans covered by this specification are intended for air circulation when located on desk tops and brackets. They are intended for multipurpose high velocity and moderately large air flow.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type, size and class of fan required (see 1.2.1).
c. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
d. Time frame required for submission of first article and number of models required (see 3.2).
e. Handles for type I, size A or B fans, if required (see 3.5.1).
f. Motor characteristics, if other than specified (see 3.5.2).
g. If suppression of electromagnetic interference is required (see 3.5.2.5).
h. Cable assembly, if other than specified and color of cable assembly, if other than specified (see 3.5.7).
i. Finish if other than manufacturer's standard practice or if the finish specified in 3.7 is desired (see 3.7).
j. Inspection responsibility, if other than specified (see 4.1).
k. Degree of preservation and degree of packing required (see 5.1 and 5.2).

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6.3 First article. Any changes or deviations of production fans from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish fans conforming to this specification.

6.4 Thrust measurement apparatus. Additional details for the thrust measurement apparatus discussed in 4.6.3.12, with description and working drawings may be obtained from the National Bureau of Standards, Washington, D.C.

6.5 Electromagnetic interference test plan. When applicable (see 3.5.2.5), the contracting officer shall arrange with the appropriate military agency (USA Communidations-Electronics Command for Army) to have the contractor's electromagnetic interference suppression test plan and test report reviewed for technical adequacy and compliance.

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 4140-A045

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-53026(ME)	Fan, Circulating

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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